

Machine Learning

CSE 343 / ECE 363

Lecture 3

Regression – Linear Models

Classification – Logistic Regression

Loss Functions: A Probabilistic View

$$\begin{aligned}\text{Error} = L(y, f(\mathbf{x}, \mathbf{w})) &= \sum_{i=1}^N (y^i - f(\mathbf{x}^i, \mathbf{w}))^2 \\ &= \sum_{i=1}^N \left(y^i - w_0 - \sum_{j=1}^d x_j^i w_j \right)^2\end{aligned}$$

$$y^i = f(\mathbf{x}^i, \mathbf{w}) = \mathbf{w}^\top \mathbf{x}^i$$

$$f(\mathbf{x}^i, \mathbf{w}) = w_1 x_1^i + w_2 x_2^i + \dots$$

The Model

- If we model the noise as a zero mean **Gaussian** random variable with variance σ^2 , the distribution is:

$$\begin{aligned}y^i &= f(\mathbf{x}^i, \mathbf{w}) + \epsilon^i = \mathbf{w}^\top \mathbf{x}^i + \epsilon^i \\ &= w_1 x_1^i + w_2 x_2^i + \dots + \epsilon^i \\ \Rightarrow \epsilon^i &= y^i - \mathbf{w}^\top \mathbf{x}^i\end{aligned}$$

$$p(\epsilon^i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\epsilon^i)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2} \left(\frac{y^i - \mathbf{w}^\top \mathbf{x}^i}{\sigma}\right)^2\right]$$

Loss Functions: A Probabilistic View

- Assuming i.i.d. errors, the joint probability of $p(\epsilon) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N)$

$$\begin{aligned} p(\epsilon) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N) &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\epsilon^i)^2}{2\sigma^2}\right) \\ &= \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp\left(-\frac{1}{2} \left(\frac{\sum_{i=1}^N (y^i - \mathbf{w}^\top \mathbf{x}^i)^2}{\sigma^2} \right)\right) \\ &= \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp\left(-\frac{1}{2} \left(\frac{\|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|}{\sigma} \right)^2\right) \quad \text{where,} \end{aligned}$$

$\mathbf{X} = [\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N] \in \mathbb{R}^{d \times N}$
 $\mathbf{y} = [y^1, y^2, \dots, y^N]^\top \in \mathbb{R}^N$

- We can view this joint probability as a function of the parameters

$$\ell(\mathbf{w}) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N | \mathbf{w}) = \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp\left(-\frac{1}{2} \left(\frac{\|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|}{\sigma} \right)^2\right)$$

This is the likelihood function

Loss Functions: A Probabilistic View

$$y^i = f(\mathbf{x}^i, \mathbf{w}) = \mathbf{w}^\top \mathbf{x}^i$$

$$f(\mathbf{x}^i, \mathbf{w}) = w_1 x_1^i + w_2 x_2^i + \dots$$

The Model

$$\text{Error} = L(y, f(\mathbf{x}, \mathbf{w})) = \sum_{i=1}^N |y^i - f(\mathbf{x}^i, \mathbf{w})|$$

- If we model the noise as a zero mean **Laplacian** random variable with scale parameter b , the distribution is:

$$\begin{aligned} y^i &= f(\mathbf{x}^i, \mathbf{w}) + \epsilon^i = \mathbf{w}^\top \mathbf{x}^i + \epsilon^i \\ &= w_1 x_1^i + w_2 x_2^i + \dots + \epsilon^i \\ \implies \epsilon^i &= y^i - \mathbf{w}^\top \mathbf{x}^i \end{aligned}$$

$$p(\epsilon^i) = \frac{1}{2b} \exp\left(-\frac{|\epsilon^i|}{b}\right) = \frac{1}{2b} \exp\left(-\frac{|y^i - \mathbf{w}^\top \mathbf{x}^i|}{b}\right)$$

Loss Functions: A Probabilistic View

- Assuming i.i.d. errors, the joint probability of $p(\epsilon) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N)$

$$\begin{aligned} p(\epsilon) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N) &= \prod_{i=1}^N \frac{1}{2b} \exp\left(-\frac{|\epsilon^i|}{b}\right) \\ &= \frac{1}{(2b)^N} \exp\left(-\frac{1}{b} \sum_{i=1}^N |y^i - \mathbf{w}^\top \mathbf{x}^i|\right) \\ &= \frac{1}{(2b)^N} \exp\left(-\frac{\|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|_1}{b}\right) \end{aligned} \quad \begin{aligned} \mathbf{X} &= [\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N] \in \mathbb{R}^{d \times N} \\ \mathbf{y} &= [y^1, y^2, \dots, y^N]^\top \in \mathbb{R}^N \end{aligned}$$

- We can view this joint probability as a function of the parameters

$$\ell(\mathbf{w}) = p(\epsilon^1, \epsilon^2, \dots, \epsilon^N \mid \mathbf{w}) = \frac{1}{(2b)^N} \exp\left(-\frac{\|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|_1}{b}\right)$$

This is the likelihood function

Maximum Likelihood

- Maximize the likelihood over all available samples

$$\hat{\mathbf{w}} = \arg \max_{\mathbf{w}} p(\epsilon^1, \epsilon^2, \dots, \epsilon^N | \mathbf{w}) = \arg \max_{\mathbf{w}} \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp \left(-\frac{\|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|^2}{2\sigma^2} \right)$$

- Since log is a monotonic function, often log-likelihood is used

$$\hat{\mathbf{w}} = \arg \max_{\mathbf{w}} \log p(\epsilon^1, \epsilon^2, \dots, \epsilon^N | \mathbf{w}) = \arg \max_{\mathbf{w}} \left(-\sum_{i=1}^N (y^i - \mathbf{w}^\top \mathbf{x}^i)^2 + \text{const} \right).$$

Measurements/data Parameters/model

Maximum (log-)Likelihood

- Valid for an **arbitrary distribution**

$$\begin{aligned}\hat{\boldsymbol{w}} &= \arg \max_{\boldsymbol{w}} \log p(\epsilon_1, \epsilon_2, \dots, \epsilon_N | \boldsymbol{w}) \\ &= \arg \max_{\boldsymbol{w}} \sum_{i=1}^N \log p(\epsilon_i | \boldsymbol{w})\end{aligned}$$

Logistic Regression

Logistic Regression

- Basic Idea:
 - Logistic regression is the type of regression we use for a binary (or discrete) target variable ($Y \in \{0,1\}$)
 - Linear regression is the type of regression we use for a continuous, normally distributed target ($Y \in \mathbb{R}^m$) variable
- Use a function to map real numbers to $\{0,1\}$
 - Thresholding
 - Some form of 'squishing' operation

Probabilistic Models

- Instead of modelling y^i as a Gaussian r.v. (as in Linear Regression),
 - model y^i as a Bernoulli random variable.
 - or more generally, as a binomial random variable.

Bernoulli and Binomial Distributions

- Bernoulli – one trial for a binary experiment

$$\begin{aligned} Pr(y|x; p) &= \begin{cases} p, & y = 1 \\ 1 - p, & y = 0 \end{cases} \\ &= p^y (1 - p)^{(1-y)} \end{aligned}$$

- Binomial – n trials of the same binary experiments
 - Bernoulli is a special case of Binomial with n=1

$$Pr(Y = y|x; n, p) = \binom{n}{y} p^y (1 - p)^{n-y}$$

Why model like this?

when x_i are discrete/mixed/grouped

- Example: x = age group of a person; y = # persons who voted
 - many individuals may get mapped to the same x

Why can't we use Linear Regression to model binary responses?

- The response (Y) is NOT normally distributed
- The variability of Y is NOT constant

Recall that Linear Regression assumes a Normal distribution with fixed variance

- Variance of Y depends on the expected value of Y
- For a $Y \sim \text{Binomial}(n, p)$ we have $\text{Var}(Y) = np(1-p)$ which depends on the expected response, $E(Y) = np$

- The model must produce predicted/fitted probabilities that are between 0 and 1
 - Linear regression produce fitted responses that vary from $-\infty$ to ∞

Linear vs. Logistic Regression

- Linear Regression models try to predict the mean of Y , i.e., $f(x;w)$
- Logistic Regression models try to predict 'p', the probability of a trial resulting in a success ($Y=1$)
 - $p \in [0,1]$ (better than $\{0,1\}$)
 - What if we try to model some invertible function of p ?
- What function of 'p' should we model?

Probability and Odds of an Event

$$\begin{aligned}\text{odds}(Y=1) &= \frac{P(Y=1)}{P(Y=0)} = \frac{P(Y=1)}{1-P(Y=1)} \\ &= \frac{p}{1-p}\end{aligned}$$

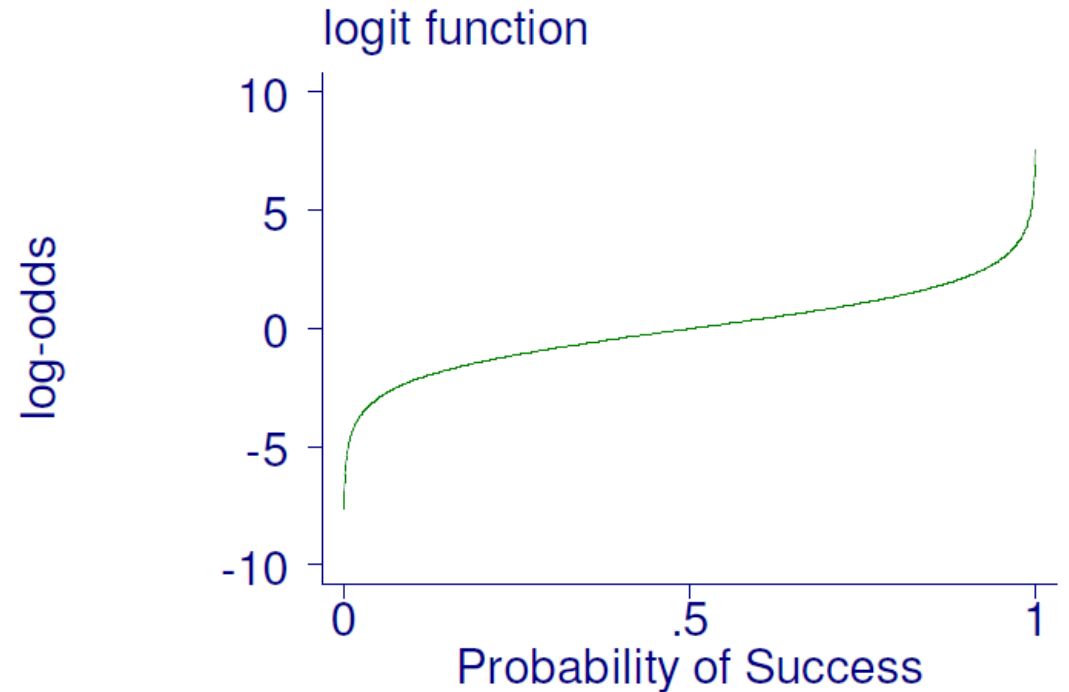
- We can go back and forth between odds and probabilities:

$$\text{Odds} = \frac{p}{1-p}$$

$$p = \text{odds}/(\text{odds}+1)$$

Logit Function (or log(odds))

- Why log(odds)?
 - Odds map 'p' from $[0,1]$ to $[0,\infty]$
 - Linear Regression predicts in the range $[-\infty, +\infty]$
 - log(odds) maps odds from $[0,\infty]$ to $[-\infty, +\infty]$
 - So we can use linear regression to predict log(odds)



Logistic Regression

- Same as Linear Regression on 'log(odds)'

$$\begin{aligned}\log(\text{odds}) &= w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d = \mathbf{w}^\top \mathbf{x} \\ \text{logit}(p) &= \log\left(\frac{p}{1-p}\right) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d = \mathbf{w}^\top \mathbf{x}\end{aligned}$$

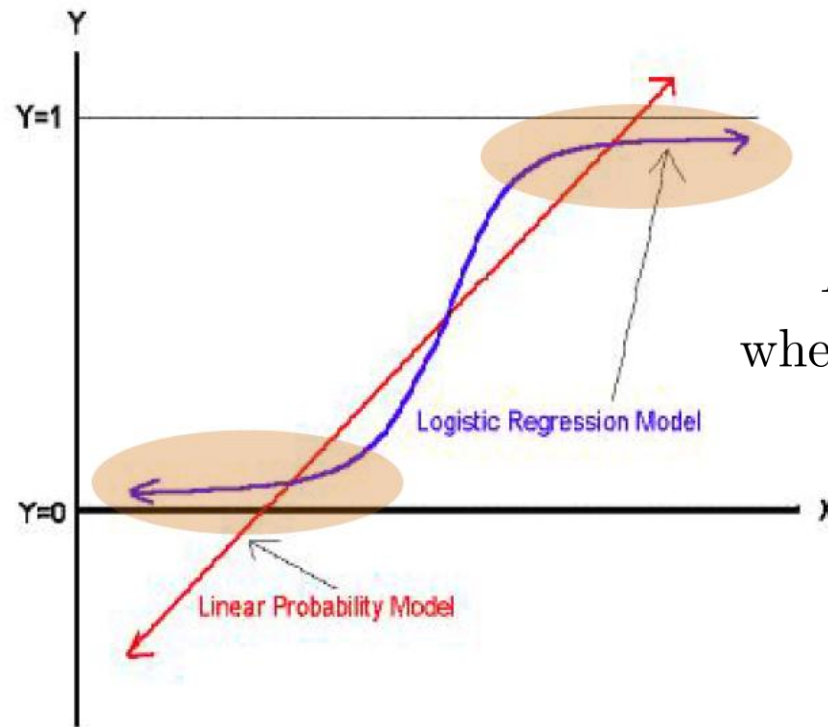
$$\begin{aligned}\left(\frac{p}{1-p}\right) &= \exp(\mathbf{w}^\top \mathbf{x}) \\ &= e^z\end{aligned}$$

- Recovering 'p=Pr(Y=1)'

$$\begin{aligned}p &= \frac{\exp(\mathbf{w}^\top \mathbf{x})}{1 + \exp(\mathbf{w}^\top \mathbf{x})} = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}} \\ 1 - p &= \frac{1}{1 + \exp(\mathbf{w}^\top \mathbf{x})} = \frac{1}{1 + e^z} = \frac{e^{-z}}{1 + e^{-z}}\end{aligned}$$

Sigmoid function $\sigma(z) : \mathbb{R} \Rightarrow [0, 1]$

Logistic regression vs. Linear regression



$$P(Y = 1|X = \mathbf{x}) = \frac{1}{1+e^{-z}}$$

where $z = \mathbf{w}^\top \mathbf{x}$

MLE

$$\begin{aligned} Pr(y|x; p) &= \begin{cases} p, & y = 1 \\ 1 - p, & y = 0 \end{cases} \\ &= p^y (1 - p)^{(1-y)} \end{aligned}$$

$$\begin{aligned} \ell(\mathbf{w}) &= \sum_{i=1}^N y^i \log p(\mathbf{x}^i; \mathbf{w}) + (1 - y^i) \log (1 - p(\mathbf{x}^i; \mathbf{w})) \\ &= \sum_{i=1}^N y^i \log \frac{p(\mathbf{x}^i; \mathbf{w})}{1 - p(\mathbf{x}^i; \mathbf{w})} + \log \left(\frac{1}{1 + \exp(\mathbf{w}^\top \mathbf{x}^i)} \right) \\ &= \sum_{i=1}^N y^i \mathbf{w}^\top \mathbf{x}^i - \log(1 + \exp(\mathbf{w}^\top \mathbf{x}^i)) \end{aligned}$$

MLE for Linear Regression

$$\begin{aligned} L(y, f(\mathbf{x}, \mathbf{w})) &= \sum_{i=1}^N (y^i - f(\mathbf{x}^i, \mathbf{w}))^2 \\ &= \|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|_2^2 \end{aligned}$$

$$\hat{y}^i = f(\mathbf{x}^i, \mathbf{w}) = \mathbf{w}^\top \mathbf{x}^i$$

$$\hat{y}^i = f(\mathbf{x}^i, \mathbf{w}) = w_0 x_0^i + w_1 x_1^i + w_2 x_2^i + \dots$$

x_j^i : j^{th} element of the i^{th} sample vector $\mathbf{x}^i$

Usually, $x_0^i = 1$

Maximum Likelihood Estimation

$$\begin{aligned}\frac{\partial}{\partial w_j} \ell(\mathbf{w}) &= \frac{\partial}{\partial w_j} \left(\sum_{i=1}^N y^i \mathbf{w}^\top \mathbf{x}^i - \log(1 + \exp(\mathbf{w}^\top \mathbf{x}^i)) \right) \\ &= \sum_{i=1}^N x_j^i \underbrace{(y^i - p(\mathbf{x}^i; \mathbf{w}))}_{\text{Prediction error in probability}}\end{aligned}$$

Prediction error in probability

Gradient ascent

$$w_j(t + 1) = w_j(t) + \eta \sum_{i=1}^N x_j^i (y^i - p(\mathbf{x}^i; \mathbf{w}))$$

- Iteratively updating the weights in this fashion increases likelihood each round.
- We eventually reach the maximum
- We are near the maximum when changes in the weights are small.
- Thus, we can stop when the sum of the absolute values of the weight differences is less than some small number.

Summary

- Logistic regression gives us a framework in which to model binary outcomes
- Uses the structure of linear models, with outcomes ($\log(\text{odds})$) modelled as a function of input variables
- Many concepts carry over from linear regression
 - logistic regression
 - Kernelization (using non-linear functions of input variables)